

Noncommuting Exponential Spectra in an Operator Algebra

Independent research note

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Abstract

Klaja and Ransford asked whether there are a Banach space E and bounded operators S, T on E for which the nonzero exponential spectra of ST and TS differ. We give a positive answer, with E separable. The construction combines their explicit matrix functions on $\mathbb{S}^4$ with Motakis's Banach space whose Calkin algebra is $C(\mathbb{S}^4)$. The crucial extra feature is that every Lipschitz scalar function has an exact multiplicative bounded lift. Matrix amplification lifts the Klaja–Ransford pair: the Calkin quotient detects that one product is not in the identity component, while the reversed product is a single explicit operator exponential.

Packet status

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Source	H. Klaja and T. Ransford, <i>Non-commutativity of the exponential spectrum</i> , arXiv:1510.08109, Question 4.2, PDF page 7 [1].
Classification	Candidate full result, likely valid subject to human mathematical review.
Novelty	Moderate confidence. The proof is a new synthesis of two published constructions; no prior connection was located in a bounded search through August 17, 2026.

As remarked just before Theorem [1.1](#), the Banach algebra $C(\mathbb{S}^4, M_2(\mathbb{C}))$ is actually a C^* -algebra. However, it is not a von Neumann algebra. Indeed [\(2\)](#) always holds in a von Neumann algebra, since the set of invertible elements is connected [\[1\]](#), Proposition 5.29 and Corollary 5.30]. This leads us to speculate whether it is possible to find counterexamples to [\(2\)](#) within other standard classes of Banach algebras. In particular, the following question seems natural.

Question 4.2. Do there exist a Banach space E and operators S, T on E such that $\epsilon(ST) \setminus \{0\} \neq \epsilon(TS) \setminus \{0\}$?

For a unital complex Banach algebra A , let $\text{Exp}(A)$ denote the connected component of 1 in $\text{Inv}(A)$, equivalently the finite products of exponentials, and put

$$\epsilon(x) = \{\lambda \in \mathbb{C} : \lambda 1 - x \notin \text{Exp}(A)\}.$$

The source asks whether $A = \mathcal{L}(E)$ can contain S, T with $\epsilon(ST) \setminus \{0\} \neq \epsilon(TS) \setminus \{0\}$.

1 Proof intuition

Klaja and Ransford already construct smooth 2×2 matrix functions a, b on $\mathbb{S}^4$ for which $1 - 2ab$ is invertible but not null-homotopic, whereas $1 - 2ba$ is null-homotopic. A representation of this function algebra by operators is not enough: a path may appear after embedding into the much larger algebra $\mathcal{L}(E)$.

The remedy is to combine a lift with a quotient detector. Motakis constructs a separable space X with $\text{Cal}(X) \cong C(\mathbb{S}^4)$ and bounded diagonal lifts $f \mapsto D_f$ for every Lipschitz f ; on Lipschitz functions this lift is exactly multiplicative. On $E = X \oplus X$, entrywise amplification lifts the smooth matrix functions, while the Calkin quotient maps them back to the originals. Thus the hard component cannot disappear. For the easy component we do not need to lift an arbitrary homotopy: its scalar entry has the explicit Lipschitz logarithm $i(\pi - 4 \arctan t)$, so its operator lift is one exponential.

2 The Lipschitz Calkin lift

We use the following consequence of Motakis's Propositions 4.1 and 4.3 and Corollary 5.3 [2].

Lemma 1 (Motakis lift). *Let K be a compact metric space. There are a separable complex Banach space X , a Schauder basis $(d_\gamma)_{\gamma \in \Gamma}$, and a map $\kappa : \Gamma \rightarrow K$ such that:*

- (i) *for every scalar Lipschitz function f on K , the rule*

$$D_f d_\gamma = f(\kappa(\gamma)) d_\gamma$$

extends to a bounded operator on X ;

- (ii) *$f \mapsto D_f$ is a unital algebra homomorphism on $\text{Lip}(K)$;*
- (iii) *if $q_X : \mathcal{L}(X) \rightarrow \text{Cal}(X)$ is the quotient map, then $f \mapsto q_X(D_f)$ extends to a Banach algebra isomorphism $\Psi : C(K) \rightarrow \text{Cal}(X)$.*

Take $K = \mathbb{S}^4$ and this space X , and put $E = X \oplus X$ (with, for example, the maximum norm). Block matrices give canonical Banach algebra identifications

$$\mathcal{L}(E) = M_2(\mathcal{L}(X)), \quad \mathcal{K}(E) = M_2(\mathcal{K}(X)), \quad \text{Cal}(E) = M_2(\text{Cal}(X)). \quad (1)$$

The compact identity follows because a finite block operator is compact if and only if each entry is compact.

For a matrix-valued function $F = (f_{ij})$ whose entries are Lipschitz, define

$$J(F) := (D_{f_{ij}})_{i,j=1}^2 \in \mathcal{L}(E).$$

Lemma 1(ii) implies

$$J(FG) = J(F)J(G), \quad J(I_2) = I_E. \quad (2)$$

Under (1), matrix amplification of Ψ^{-1} defines an isomorphism

$$\Theta : \text{Cal}(E) \longrightarrow M_2(C(K))$$

such that

$$\Theta(q_E(J(F))) = F. \quad (3)$$

If both F and F^{-1} have Lipschitz entries, then (2) also gives $J(F)^{-1} = J(F^{-1})$.

3 The operator counterexample

Write a point of $\mathbb{S}^4$ as $(z_0, z_1, t) \in \mathbb{C}^2 \times \mathbb{R}$, where $|z_0|^2 + |z_1|^2 + t^2 = 1$. Define the smooth matrix functions

$$a(z_0, z_1, t) = \frac{1}{1+it} \begin{pmatrix} z_0 & 0 \\ z_1 & 0 \end{pmatrix}, \quad b(z_0, z_1, t) = \frac{1}{1+it} \begin{pmatrix} \bar{z}_0 & \bar{z}_1 \\ 0 & 0 \end{pmatrix}. \quad (4)$$

Klaja and Ransford prove that

$$c := I_2 - 2ab : \mathbb{S}^4 \longrightarrow GL_2(\mathbb{C}) \quad (5)$$

is not null-homotopic [1, Theorems 1.1 and 1.2]. Equivalently,

$$c \notin \text{Exp}(C(\mathbb{S}^4, M_2(\mathbb{C}))). \quad (6)$$

All entries of a, b, c are Lipschitz. Since c is a smooth map into $GL_2(\mathbb{C})$ on a compact manifold, c^{-1} also has Lipschitz entries.

Theorem 1. *There exist a separable complex Banach space E and operators $S, T \in \mathcal{L}(E)$ such that*

$$\frac{1}{2} \in \epsilon(ST) \quad \text{and} \quad \frac{1}{2} \notin \epsilon(TS).$$

Consequently Question 4.2 of Klaja–Ransford has a positive answer.

Proof. With $E = X \oplus X$ as above, set

$$S = J(a), \quad T = J(b).$$

Equations (2) and (5) give

$$I_E - 2ST = J(c). \quad (7)$$

This operator is invertible, with inverse $J(c^{-1})$. If it belonged to $\text{Exp}(\mathcal{L}(E))$, then applying the unital quotient homomorphism q_E and the isomorphism Θ would, by (3), put c in $\text{Exp}(C(\mathbb{S}^4, M_2(\mathbb{C})))$, contradicting (6). Hence

$$I_E - 2ST \notin \text{Exp}(\mathcal{L}(E)). \quad (8)$$

For the reversed product, direct multiplication in (4) gives

$$I_2 - 2ba = \begin{pmatrix} \phi(t) & 0 \\ 0 & 1 \end{pmatrix}, \quad \phi(t) = -\left(\frac{1-it}{1+it}\right)^2. \quad (9)$$

For $-1 \leq t \leq 1$, put

$$g(t) = i(\pi - 4 \arctan t).$$

The identity

$$\frac{1-it}{1+it} = e^{-2i \arctan t}$$

shows that $\phi = e^g$. In particular g and ϕ are Lipschitz. If $G = \text{diag}(g, 0)$, then (2), or simply evaluation on every basis vector, yields

$$I_E - 2TS = J(\text{diag}(\phi, 1)) = \exp(J(G)) \in \text{Exp}(\mathcal{L}(E)). \quad (10)$$

Finally,

$$\frac{1}{2}I_E - ST = \frac{1}{2}(I_E - 2ST), \quad \frac{1}{2}I_E - TS = \frac{1}{2}(I_E - 2TS).$$

Since $\frac{1}{2}I_E = \exp(-\log(2)I_E)$ belongs to $\text{Exp}(\mathcal{L}(E))$ and $\text{Exp}(\mathcal{L}(E))$ is a subgroup, multiplication by this scalar preserves membership in $\text{Exp}(\mathcal{L}(E))$. Statements (8) and (10) therefore give the two asserted exponential-spectrum memberships. $\square$

4 Verification, scope, and novelty

The proof uses only exact algebraic identities after the two cited constructions. The finite direct-sum Calkin identity (1), multiplicativity (2), quotient detection (3), regularity of c^{-1} , the orientation of ST versus TS , and the logarithm in (9)–(10) were audited separately. No numerical computation is used.

The result settles Question 4.2 as stated and strengthens it by making E separable. It does not settle the adjacent Question 4.1, which asks for a universal torsion consequence for arbitrary Banach algebras. The space E is obtained from a Bourgain–Delbaen construction and is not claimed to be a classical sequence or function space.

A bounded novelty search covered the run indexes, the local arXiv corpus, exact question text, author/title combinations, Calkin realizations, and a citation list for [1]. The closest later result found is Daniel–Ghosh [3], which proves commutativity in $\mathcal{L}(\ell^p \oplus \ell^q)$ and states that Calkin commutativity implies operator-algebra commutativity. It does not give the existence theorem above and does not cite Motakis. No source connecting [1] and [2] was located. Because the ingredients are published even though their combination appears new, novelty confidence is moderate.

Human review should focus on (1)–(3) and the explicit logarithm in (10).

References

- [1] H. Klaja and T. Ransford, *Non-commutativity of the exponential spectrum*, J. Funct. Anal. 272 (2017), 4158–4164; arXiv:1510.08109.
- [2] P. Motakis, *Separable Spaces of Continuous Functions as Calkin Algebras*, arXiv:2110.10868.
- [3] S. Daniel and A. Ghosh, *A note on commutativity of the exponential spectrum in the operator algebra $\mathcal{B}(\ell^p \oplus \ell^q)$* , Acta Math. Hungar. 177 (2025), 280–296, doi:10.1007/s10474-025-01571-x.